CSH interview: final preparation pack

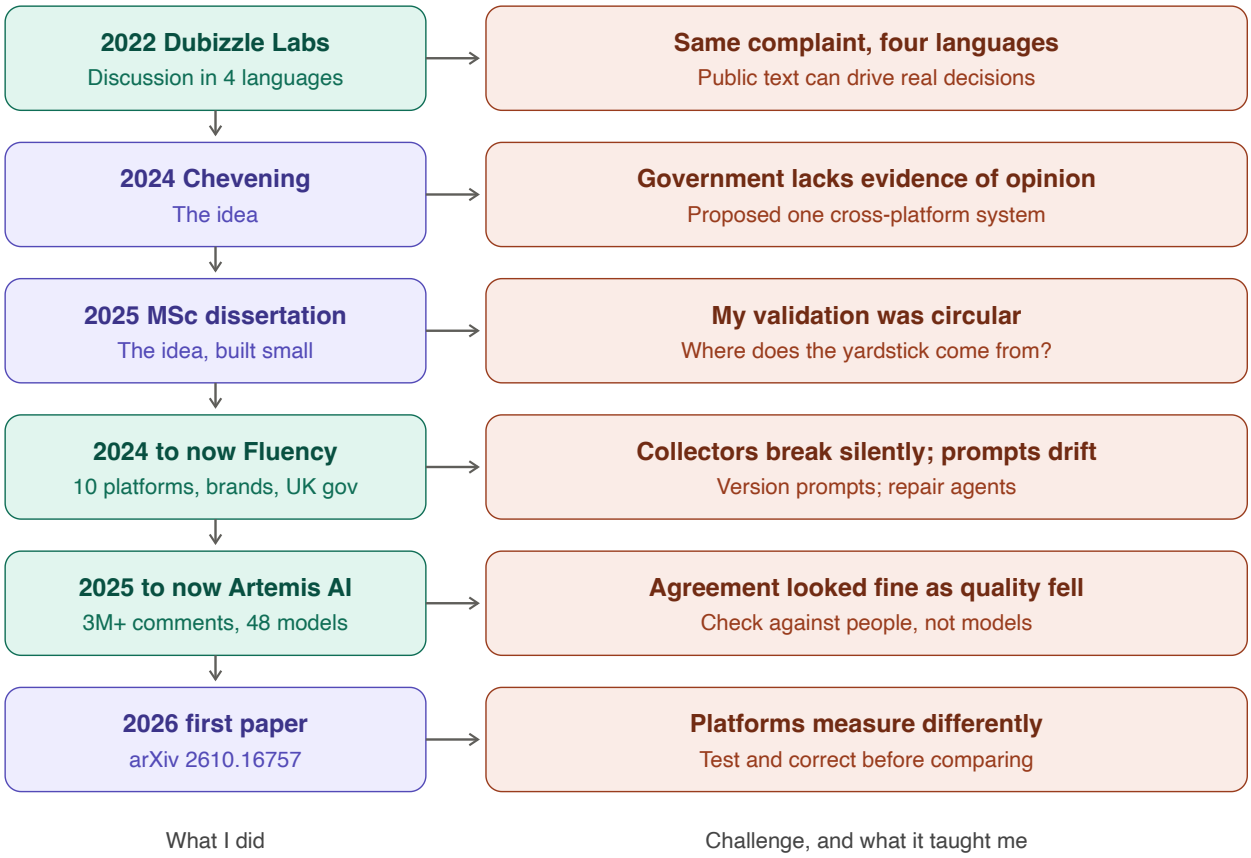
Preliminary interview **Tue 20 Oct 2026, 4:40–4:55 pm UK = 8:40 pm Lahore** · Zoom 843 5332 6447 · Graduate Program in Complexity Science, Digital Innovation School · interviewer contact: Carrie Cowan, Head of DIS

1. How this interview works (researched)

- **Format:** about 10 minutes of standard questions on your prior experience and future interests, plus a few minutes of instructions; in English on Zoom. CSH does not share questions in advance and says there is no need to prepare or study.
- **It counts:** the recording and transcript become part of your application package and are read by the Selection Committee alongside your written answers. So keep every fact consistent with what you submitted.
- **What the committee looks for** (CSH FAQ): fit between your background and CSH research, and "potential for independent and rigorous scholarship in the area of complexity science".
- **After this:** Stage 1 committee review → Stage 2 panel interview on Zoom, 7–14 January 2027, where you present a paper chosen by the committee and answer questions on it → Stage 3 finalists present their research ideas at CSH, 15–16 February → offers by mid-March. Referee letters must arrive by **1 November**.
- **What good answers look like** (PhD interview guides): specific, concrete examples of what you did; a clear reason for a PhD that comes from a real experience, not a slogan; naming their work that moved you; and showing gaps as things you are already learning.

Three things to land in any answer.
Question: do machine and LLM labels of online text mean the same thing across platforms, models and time, and how do we correct comparisons when they do not?
Proof: your first paper, with Dr Ella Haig, on arXiv since 14 October (2610.16757), which grew from your dissertation's circular validation.
Edge: eight years collecting and labelling social media at scale (Dubizzle, Fluency, Artemis AI); data is never your bottleneck.

2. Your story: how your social media work grew, step by step



Teal = industry, purple = academic and ideas. Every step was about social media data, and each challenge led to the next step.

1. Dubizzle Labs (2022–2024): the first time public text drove decisions. You designed the company's first web data system across 15 markets and led three engineers. Its dashboards combined listings with what people said online, in English, Arabic, Urdu and Hindi, to show what customers wanted and to predict revenue for decision-makers. *Challenge:* the same complaint looks completely different in four languages and on different sites. *What it taught you:* online text can drive real decisions, so it had better be measured well. You were promoted twice.

For this PhD: multilingual text is the norm in CSH work on narratives and migration discourse; you already know how differently the same view reads across languages, which is exactly the kind of non-equivalence your PhD would test.

2. Chevening (2024): the idea. You won the UK Government's Chevening Scholarship (selected from 70,000+ applicants) for helping students and junior colleagues into freelance and remote careers, and for a proposal: a centralised, cross-platform social media analytics system that would give government evidence of public opinion for decisions. *That proposal is the seed of everything after it.*

For this PhD: it shows the question is long-standing and policy-facing, not a fashion: comparable evidence of public opinion across platforms is what CSH's narratives and polarisation projects also need.

3. MSc dissertation (2025): the idea, built at small scale. At Portsmouth you built a version of the Chevening idea: public reaction to UK economic policy across 255,446 comments from X, Reddit, YouTube and Quora, with BERT, baselines, significance tests and forecasting. *Challenge:* you found your own validation was circular (labels seeded by VADER, checked against VADER), and you had assumed a "negative" post means the same on every platform. *What it taught you:* always ask where the yardstick comes from, and test that an instrument works the same in each group before comparing.

For this PhD: this is the research habit a PhD runs on: finding the flaw in your own work and turning it into a question. It is also the starting point of your proposed project.

4. M+C Saatchi Fluency (Dec 2024 to now): social media at professional scale. Ten platforms collected every day for global brands and UK government communications teams; analyst on a twelve-month

study of about 271,000 mentions with a topic codebook and LLM tagging. *Challenges:* collectors break silently when sites change, which quietly thins a time series; and an LLM's labels change whenever its prompt changes. *What you did:* agents that detect and repair broken collectors, with a person approving each fix; every LLM tag stamped with its prompt version; prompts piloted three times on the same rows before a full run.

For this PhD: you would build reproducible, versioned corpora for the narratives-at-scale project from day one, and you know from practice that LLM labels are instruments that must be recorded and tested.

5. Artemis AI (2025 to now): building your own social media product. As co-founder and CTO you lead the technology in a team of eight: Meta's official API, more than 3 million Facebook and Instagram comments with images and video, an LLM labelling chain feeding your own classifiers, and 48 models planned for the December beta, run from an admin panel with a 1,438-ticket tracker, an accuracy tracker for the NLP models, a QA dashboard that asks "are the models still accurate and did the data arrive?", and alerts for checks that have quietly stopped running. The real challenges were measurement problems:

- **The agreement trap.** When you compared two models, their agreement went *up* on three of five tasks while both were collapsing onto the most common answer. Agreement looked healthy exactly when quality was failing. Lesson: test against fixed human-checked labels, never against another model.

  For this PhD: validating LLM labels against people is central to Alexander Hoyle's work and to any study that uses LLMs as annotators.
- **Silent drift.** Agreement on post sentiment fell from 77% to 37% within weeks, and tracing it showed how a labelling setup can drift without anyone noticing. Lesson: monitor labels continuously and version everything.

  For this PhD: LLM annotators change version during a study; you know how to detect and handle that, which is one facet (time and model) in your proposed project.
- **What counts as "normal"?** To define a comment storm, the team screened 207,192 comments on 369 unusually busy posts from February to July and found almost no real storms aimed at the page. Lesson: the hard part is each page's baseline, not the alarm.

  For this PhD: measuring polarisation or hostility needs a baseline per community and platform; this is the same logic as comparing commenters with watchers in Chen et al.
- **Banter and dialect.** Friendly banter and dialect get misread as toxicity, so you built filters for them: the same words mean different things in different communities.

  For this PhD: minority and migration discourse (a CSH springboard project) is exactly where mislabelling a community's language would bias results.
- **Hidden bias.** Pages repost their winners, and duplicates matched only on caption text miss reposted images, which inflates what "works". Lesson: check how the data was generated before trusting a pattern.

  For this PhD: visible traces are selected; Chen et al. show the same for commenters, and a PhD has to correct for it.
- **Platform dependence.** Everything depends on Meta's API access, app review and expiring page tokens. Lesson: the platform shapes what you can see at all.

  For this PhD: platform access and affordances are part of the measurement, which is the core claim of your paper and the reason for data donation designs like Chen et al.'s.

6. Your first paper (October 2026): the lessons, made into research. With Dr Ella Haig you turned all of this into one question and a method: treat each platform like a survey mode, test whether machine labels mean the same thing on each, and correct them before comparing. In your simulation, skipping that step points the overall mood the wrong way on nearly a quarter of days.

For this PhD: the paper is the first chapter: the protocol and simulation are ready, so the PhD can move straight to new data, LLM annotators, time, and network diffusion.

The 60-second version for "Tell us about yourself": "My work has always been about social media data, and it has grown one step at a time. At Dubizzle I first saw public online text drive real business decisions, in four languages. That led to my Chevening proposal: a cross-platform system to give government evidence of public opinion. I built a small version of it in my MSc dissertation, and found my own validation was circular. At Fluency I work with social media at professional scale for brands and UK government, and at Artemis AI, my start-up, I learned that labels can look reliable exactly when they are

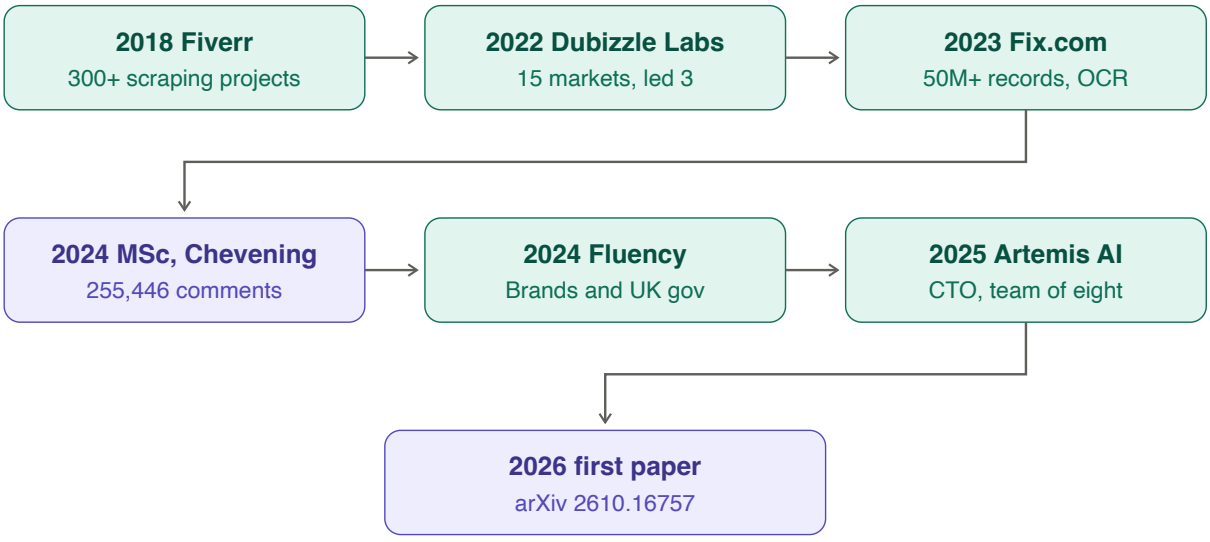
failing. My first paper, out last week, turns those lessons into a method for checking whether platforms can be compared. A PhD at CSH is the next step: doing this rigorously, with network science."

2a. What each step gives this PhD

From	You bring to the CSH PhD	Where it lands at CSH
Dubizzle	Multilingual collection and analysis at scale	Polarization and minority representation (migration discourse)
Chevening	A policy-facing question you have held for years	Evidence for public bodies; Digital Innovation School aims
MSc dissertation	Full research cycle; honest self-critique	"Independent and rigorous scholarship" (the committee's criterion)
Fluency	Reproducible LLM coding of large discourse; reliable collection	Understanding narratives at scale
Artemis AI	Validating LLM labels against people; baselines; drift	Alexander Hoyle (measurement and validation); Elisa Omodei (online social systems)
First paper	A ready protocol and simulation: chapter one	Your proposed project; extend with network diffusion
What you still need	Network science of diffusion; causal inference	Exactly what CSH teaches: the reason to do this PhD here

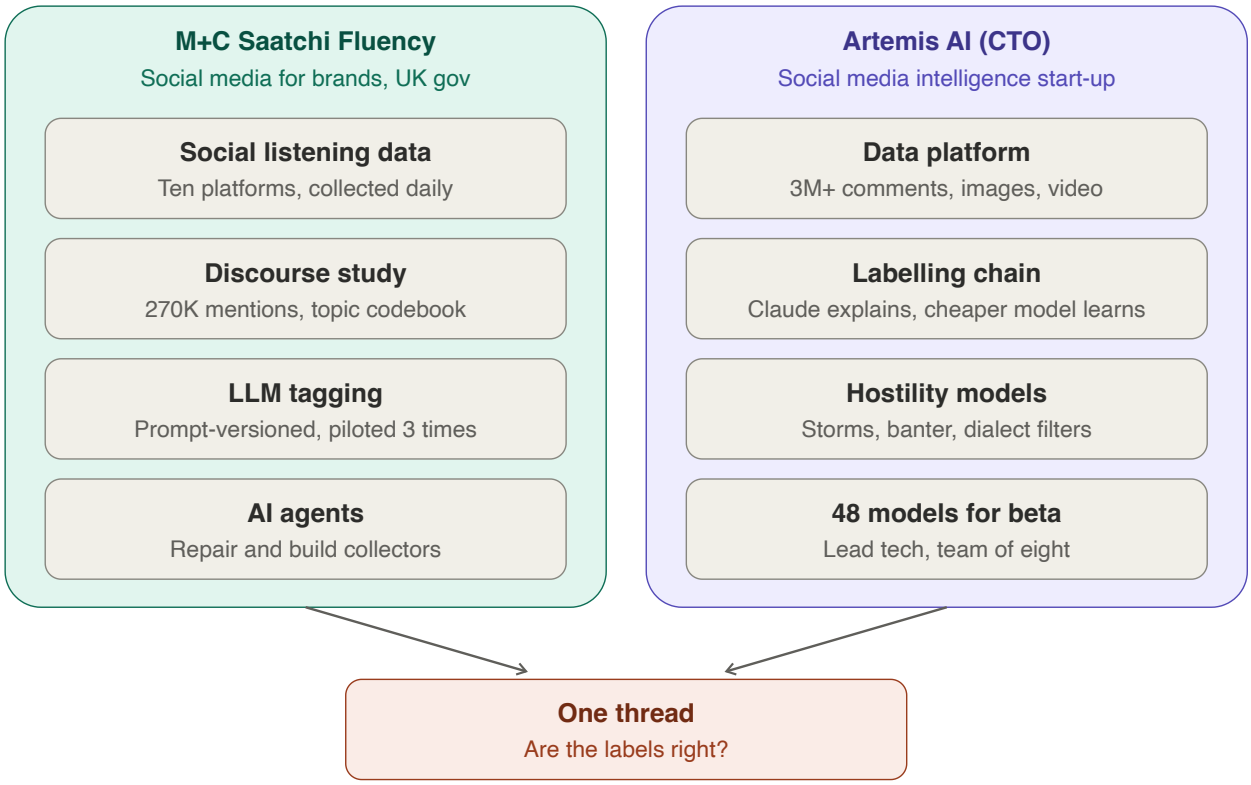
One line that ties it all to the PhD: "Everything I have done has been about measuring public opinion from social media; each step showed me a new way those measurements fail. The PhD is where I learn to test and fix them properly, and to see how corrected measures change what we conclude about narratives and polarisation."

2b. Your full path, 2018 to 2026



Teal = industry, purple = academic. Fiverr taught scraping and clients; Dubizzle multilingual data and leading people; Fix.com data quality at scale; MSc research method and honest validation; Fluency social media analysis that decisions rely on; Artemis AI building and evaluating models; the paper turned a weakness into research.

3. What you did at Fluency and Artemis AI



M+C Saatchi Fluency (London, since December 2024). Fluency is the social and web intelligence side of M+C Saatchi, and its reports go to global brands such as Amazon, Ford, Nike and Reckitt, and to UK government communications teams. Your work there has two halves.

- **Research and analysis.** You were an analyst on a twelve-month study of public discourse: 100,845 Brandwatch mentions and 170,542 Reddit mentions. Instead of hand-reading them, the team built a codebook from BERTopic topics, then used an LLM to tag every mention. The study used two research habits you now treat as standard: every tag carries a hash of the exact prompt version that produced it, so results can be traced and repeated; and the prompt was run three times on the same 500 rows before the full run, to see whether the labels were stable. In your portfolio you called this treating a prompt as an instrument whose version must be recorded. It is the closest you have come to studying narratives at scale.
- **Data and AI engineering.** You collect from ten platforms every day, feeding a governed master table. When a forum changes its layout, a crawler usually breaks silently: the series does not stop, it just thins, which quietly biases any trend. You designed a self-healing AI agent that diagnoses the break, rewrites the collector, and tests it on the live site before anything ships, plus a builder agent that creates new collectors. Both follow fixed rules, and a person approves every change.

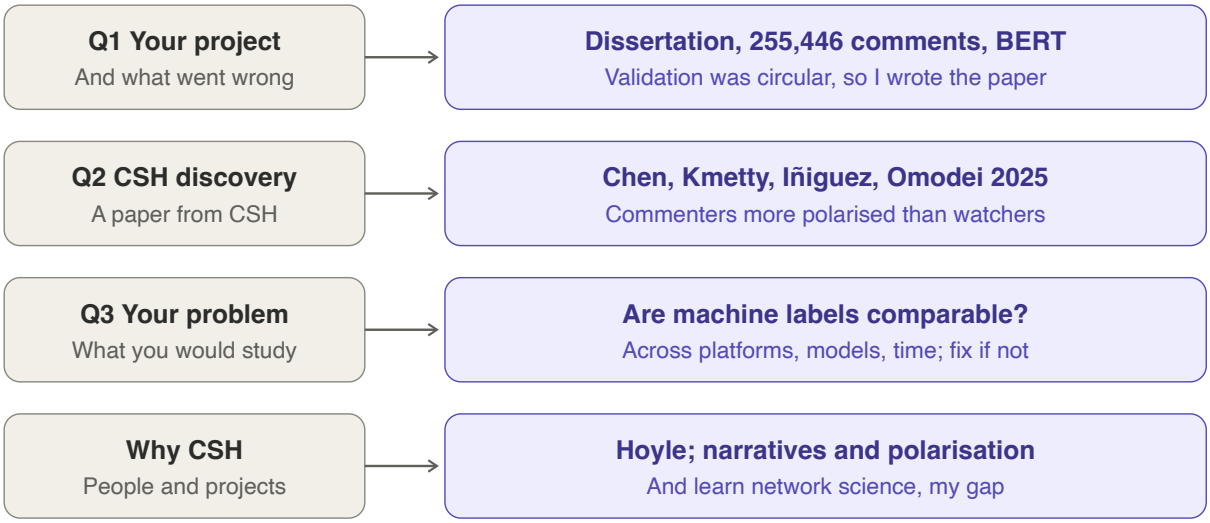
Artemis AI (co-founder and CTO, since 2025). A social media intelligence start-up helping creators and businesses understand their audiences, heading to a beta in December 2026. You lead the technology in a team of eight.

- **Data.** More than 3 million Facebook and Instagram comments on 111,000+ posts, collected through Meta's official API, with images and videos read through OCR, CLIP and Whisper.
- **Labelling, which is your research question inside a product.** Claude labels about 6,000 posts and writes a reason for each label. Rules mined from those reasons become the prompt for DeepSeek, a cheaper model, which labels about 12,000 more. Those labels train your own classifiers (XLM-R, DistilBERT, DeBERTa). The team hand-labelled about 500 posts and comments to check both models, and you are building evaluation around a human-labelled golden set so no model reaches users unchecked.
- **Hostility, measured carefully.** Toxicity and intent classifiers, a storm detector for sudden bursts of hostile comments, and filters so that dialect and friendly banter are not mistaken for abuse. That last one is a measurement problem in miniature: the same words mean different things in different communities.

- **Scale of the build.** 48 models in the agreed architecture, from comment intent to audience communities, and Art-E, an advisor that answers only from each client's own data, with citations.

Say it in one sentence: "At Fluency and Artemis AI, brands and governments act on machine-labelled numbers that are rarely checked against people; my PhD at CSH would build the methods to check them, and show what changes when we do."

4. What you told CSH (keep answers consistent)



Answer 1: a research project, your role, and what did not go as expected. Your MSc dissertation at Portsmouth (2025, supervisor Dr Ella Haig) asked how public reaction to UK economic policy differed across platforms. It was yours from question to write-up: you obtained ethics approval, collected 255,446 public comments from X, Reddit, YouTube and Quora through official APIs and licensed feeds, and built the whole analysis in Python: VADER and BiLSTM baselines, a fine-tuned BERT classifier, calibration, McNemar and bootstrap tests, chi-square tests across topics, Prophet forecasts, NRC emotions, and LIME and integrated gradients to see what the model relied on. You also hand-labelled 200 posts.

What went wrong, told honestly: BERT scored a macro-F1 of 0.878, but its training labels had been seeded by the VADER lexicon, the score was largely in-sample, and your 200-post check reviewed VADER's own labels, so the agreement (kappa about 0.78) measured agreement with VADER, not truth. The validation was circular: it showed BERT had learned to copy VADER. Underneath was a second untested assumption, that "negative" means the same on Reddit as on YouTube. *What you learned:* an agreement score describes the reference as much as the model, so the first design question is where the yardstick comes from; and a comparison between groups needs evidence that the instrument behaves alike in each, a routine test in survey research that text analysis rarely runs. That became the paper with Dr Haig, and for its application you fixed fifteen policy events in advance, before computing any counts.

Answer 2: a discovery from a CSH group. You chose Chen, Kmetty, Iñiguez and Omodei (2025), "The public that engages invisibly", *Communication Methods and Measures* 19(4): 294–312. By linking a survey of Hungarian internet users to the YouTube histories they donated, they show that people who comment on political videos are more polarised and more selective than people who only watch, so studies built on visible traces such as comments overstate how divided audiences are. *Your question:* is the gap between commenters and watchers stable enough, across platforms, topics and time, that a small donated panel could correct polarisation estimates built from public comments at scale? *Why it matters to you:* it has the same structure as your problem. Public comments are a large, cheap, selected instrument; donated data is a small, expensive, closer criterion. If the relation holds steady, a few hundred donors could calibrate millions of comments, just as a small hand-labelled sample anchors a machine-labelled corpus; if it breaks, knowing where is itself a finding. You meet the same gap at work: brands judge their pages by comments, though most of their audience never comments.

Answer 3: a problem you want to explore. *Statement:* when is a machine-labelled construct (sentiment, stance or a narrative frame) comparable across platforms, annotator models and time, and how should the differences be corrected before we track how narratives spread between platforms? *Why it matters:* research on narratives and polarisation increasingly pools platforms and uses LLMs as annotators, and any difference mixes real change with how each platform elicits expression and how each model reads it, while the models change version underneath us. You cited Gaynor, Miler, Goel, Hoyle and Resnik (2026, *Journal of Politics*):

legislators' positions differ across votes, speeches and tweets; Fan et al. (2025) on removing the medium from embeddings by concept erasure; Licht et al. (2025) on LLMs measuring scalar constructs; and Hoyle et al. (2021) on automated metrics drifting from human judgement. *Why tractable, in four steps:* (1) your simulation has known ground truth and runs in seconds, so you can add models and time as facets with no new data; (2) you can build an event-anchored corpus from official APIs, labelled by several LLMs, with a larger hand-labelled sample per platform and a second coder, then split the variance and test each frame for DIF; (3) concept erasure tests whether removing the platform signal also removes real differences; (4) corrected and uncorrected measures go into a network diffusion analysis to show when conclusions change. *For whom:* researchers who pool platforms or switch models mid-study; public bodies, NGOs and journalists reading social-listening dashboards; anyone asking whether a narrative really moved between platforms. *What you would learn at CSH:* the network science of diffusion.

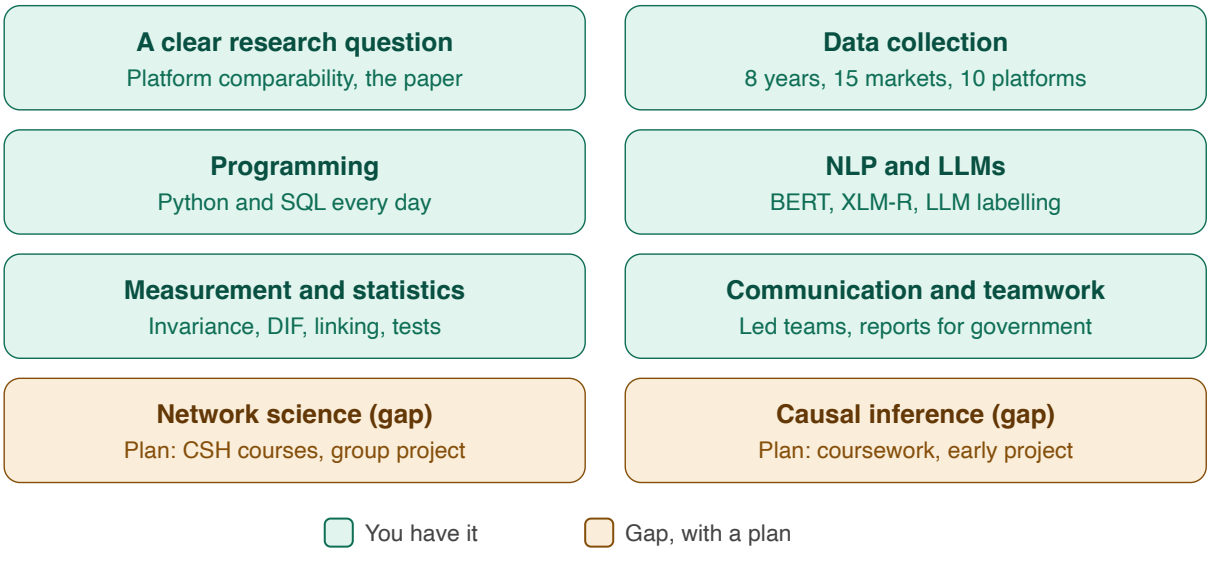
Your portfolio (sent with the CV) listed six pieces, each with why it was there: the paper; the dissertation; the Artemis labelling chain; the Fluency twelve-month study; the self-healing and builder agents ("a collector that breaks silently does not stop a time series, it thins it"); and the multilingual Dubizzle dashboards that predicted revenue and showed what customers said.

4b. The people and projects you named (verified 19 Oct)

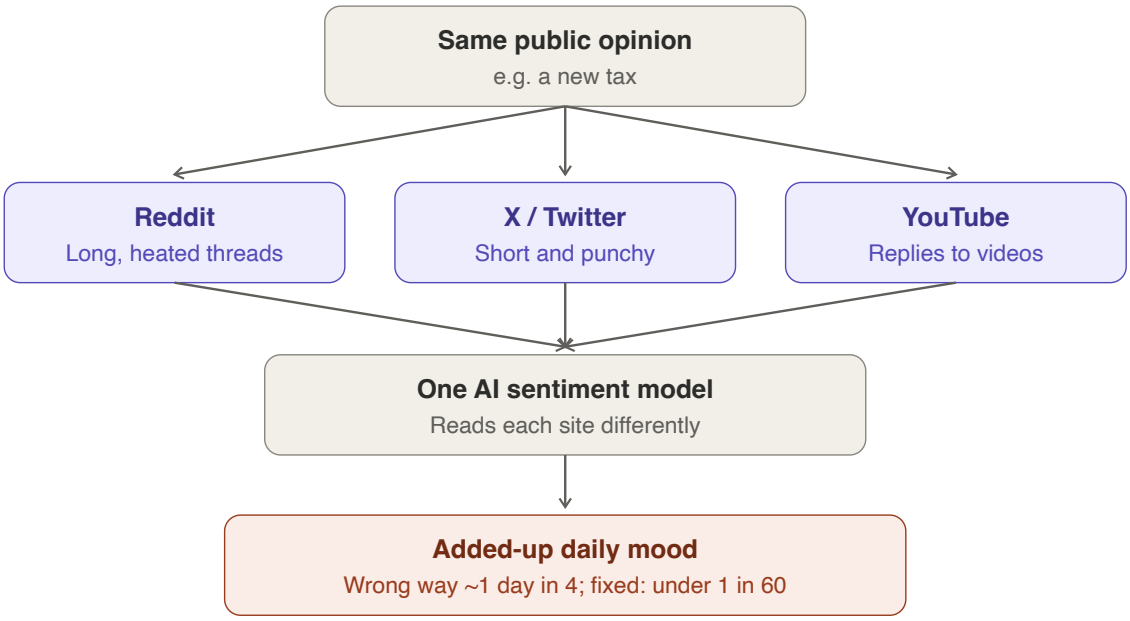
- **Alexander Hoyle**: CSH Digital Innovation School faculty, listed for NLP in social science, measurement and validation, and reasoning in LLMs; PhD Maryland 2024 with Philip Resnik, currently also associated with the ETH AI Center. His thesis is about developing and measuring latent constructs from text with human-centred validation. He is the closest match to your question; say "his work on validating NLP measures".
- **Elisa Omodei**: DIS faculty (computational social science, online social systems, biases and inequalities) and a co-author of the paper you chose for Answer 2. **Gerardo Iñiguez** is her co-author on it.
- **Fariba Karimi**: DIS faculty working on homophily and biases in networks and algorithms, relevant to polarisation and minority visibility.
- **Springboard projects** you referenced, both on CSH's 2027 list: *Understanding narratives at scale* (how narratives spread and shape discourse, with NLP, network science and large text data) and *Polarization and minority representation on online platforms* (how migration and minority-related discourse evolves online and relates to polarisation). Related ones worth knowing: *Designing link recommendation to reduce polarization* and *Identifying causal mechanisms for online behavior*.

5. What this PhD needs, and what you bring

Every strength below is something the CSH PhD uses from the first month; the two gaps are what CSH's training and faculty provide, which is why this program and not another.



6. Your paper in 20 seconds



“AI reads opinions differently on Reddit than on YouTube, because people write differently on each. So adding platforms together can flip the answer: in my simulation, the overall mood points the wrong way on nearly a quarter of days, and under one day in sixty once each platform is corrected. The paper gives a checklist to test whether a comparison is fair, and a way to fix it.”

7. Practice answers (45–75 seconds each)

Tell us about yourself.
Use the 60-second story at the end of section 2: Dubizzle, Chevening idea, dissertation, Fluency, Artemis AI, first paper, and CSH as the next step. Practise it out loud until it takes under a minute.

Describe a research project and what went wrong.
Use section 4, Q1: the project, the circular validation you found yourself, the paper it became, the quarter-of-days result, and the lesson about independent yardsticks.

What have you built in industry, and what did it teach you?
Use section 3: one Fluency example (prompt-versioned LLM tagging), one Artemis AI example (explained labels, dialect and banter filters), then the one-sentence thread.

Why a PhD, and why now?
Every job I have had showed me a new way social media measurements fail: languages at Dubizzle, my own validation in the MSc, silent drift at Fluency, and labels that look reliable while failing at Artemis AI. Industry let me find these problems but not study them properly. The paper showed me I can, and a PhD at CSH gives me the methods I lack, network science and causal inference, to finish the job.

Why CSH, and why complexity science?
Opinion online is a system, not a pile of posts: people, platforms and algorithms shape each other, and what we can see is a selected slice. Chen and colleagues showed commenters are more polarised than watchers, which is my problem in another form. Alexander Hoyle's validation work is closest to mine, and I want to learn network science and diffusion, which I don't have yet.

What would you work on?
Use section 4, Q3, in three steps: extend the simulation, build an event-anchored corpus labelled by several LLMs with human checks, then show when diffusion conclusions change after correction.

Strengths and gaps?
Use section 5: name two strengths with evidence, then the two gaps and how you will close them.

Teamwork, a setback, five years?
Team: led three at Dubizzle and three at VendueTech, lead the technology in a team of eight at Artemis AI. Setback: the circular validation. Five years: researching how online opinion is measured, so the numbers people act on are tested against people.

Questions for us? (one only)
How are students matched with supervisors in the first year, and is co-supervision across groups common?

8. On the day

- Join at **8:35 pm Lahore**; test Zoom, camera and microphone; quiet room, light in front of you, look at the camera.
- Keep this pack open but do not read from it. Answer the question, then stop. A short pause before answering is fine.
- Same facts as your application: 255,446 comments; team of eight at Artemis AI; arXiv 2610.16757; MSc Distinction, Chevening 2024/25.
- Recording is normal; only object at the start if you do not consent.

9. If you pass: Stage 2 (7–14 January)

You will present a paper chosen by the committee. Past papers: 2026 Clauset, Moore and Newman, "Hierarchical structure and the prediction of missing links in networks" (*Nature* 2008); Couzin et al., "Uninformed individuals promote democratic consensus in animal groups" (*Science* 2011); 2025 Albert, Jeong and Barabási, "Error and attack tolerance of complex networks" (*Nature* 2000); 2024 Watts and Strogatz, "Collective dynamics of small-world networks" (*Nature* 1998). CSH's advice: read carefully, google, work

through the equations or code, and be ready to explain methods, results, conclusions and critiques. Start reading these after Tuesday.

Sources: CSH Graduate Program FAQ (csh.ac.at/education/dis-faq); CSH DIS faculty list (csh.ac.at/education/dis-faculty) and springboard projects (csh.ac.at/education/dis-projects); Crossref record for doi 10.1080/19312458.2025.2542725; CSH invitation email of 14 Oct 2026; AcademicPositions, “The 5 most important questions you’ll be asked during a PhD interview”; Researcher.Life, “PhD admission interview questions and how to answer them”; your CSH answers and CV (13 Oct 2026).